

Institute of Leather Engineering and Technology
University of Dhaka

TANNERY WASTEWATER TREATMENT

Study Material for Undergraduate Students

CHAPTER 1

PRIMARY TREATMENT

Physical and Physicochemical Processes

Screening · Grit and Grease Removal · Equalisation · Neutralisation

Sulphide Oxidation · Chrome Recovery · Coagulation–Flocculation · Sedimentation

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HOW TO USE THIS CHAPTER

Sections 1.1 to 1.4 give the background you need before studying any individual unit: what tannery wastewater contains, how it is collected, and what primary treatment is meant to achieve. Read these first — the definitions given there are used throughout all three chapters and are not repeated later.

Sections 1.5 to 1.15 treat the unit operations one by one, in the order in which the wastewater actually meets them. Each follows the same pattern: purpose, principle, process description, components, operating parameters, pollutants removed, advantages, limitations, and tannery application.

Sections 1.16 to 1.18 draw the chapter together.

1.1 Introduction: Why Treatment Begins Here

Wastewater treatment is organised as a TRAIN: the water passes through a sequence of stages, and each stage removes what the previous one could not. The three families of unit operation are physical, chemical and biological, and they are applied broadly in that order.

Primary treatment is the first stage of that train. Its job is not to produce clean water — it cannot — but to remove the coarse, heavy, floating and chemically hazardous material, and to deliver a uniform, non-toxic flow to the stages that follow. A well-run primary stage typically removes 50–70% of the suspended solids and 25–40% of the BOD; the rest is left for later.

For tannery wastewater this stage carries an unusually heavy responsibility, for two reasons that will recur throughout this chapter.

- **It protects the biological stage.** Sulphide and chromium are poisons to micro-organisms. If they are not removed here, the biological treatment described in Chapter 3 cannot work at all.
- **It decides whether chromium becomes a product or a hazardous waste.** Chromium recovered from a segregated stream is a raw material worth money. The same chromium, once mixed into the general effluent, contaminates all the sludge produced by the whole plant.

KEY IDEA

Primary treatment does not clean water to a dischargeable standard. It removes what can be screened, floated and settled, destroys the chemical poisons, and evens out the flow — so that the later stages can do their work reliably.

Most failures of secondary treatment are in fact failures of primary treatment.

1.2 Tannery Wastewater: Composition and Vocabulary

Before studying any treatment unit we must know what is being treated, and we must agree on the words used to measure pollution. The definitions in this section are used in all three chapters and are not repeated later.

1.2.1 Measures of pollution load

Term	Meaning	Practical note
BOD₅	Biochemical Oxygen Demand: the oxygen micro-organisms consume in degrading the biodegradable organic matter in five days, at 20 °C (mg/L)	Measures the BIODEGRADABLE load — what biological treatment can remove
COD	Chemical Oxygen Demand: the oxygen required to oxidise ALL organic matter chemically (mg/L)	Always higher than BOD. The difference COD – BOD is roughly the non-biodegradable (refractory) fraction
TSS	Total Suspended Solids: undissolved particles retained on a filter (mg/L)	What settling and filtration remove
TDS	Total Dissolved Solids: dissolved salts passing the filter (mg/L)	Cannot be settled, filtered or biologically removed
Settleable solids	The part of TSS that settles within one hour (mL/L, Imhoff cone)	What a clarifier will actually capture
Turbidity	Cloudiness caused by suspended and colloidal particles scattering light (NTU)	A quick indicator of clarification performance
pH	Measure of acidity or alkalinity (0–14)	Tannery streams swing from about 2.5 to 12

Table 1.1 Vocabulary of pollution measurement. Keep this table beside you; the rest of the material uses these terms constantly.

A useful diagnostic follows from the first two rows. The RATIO BOD/COD indicates how biodegradable a wastewater is. A ratio above about 0.5 means the waste is readily biodegradable; below about 0.3 it is largely refractory and biological treatment alone will not be sufficient. Tannery effluent typically falls between 0.3 and 0.5 — partly biodegradable, which is why it needs both biological and physicochemical treatment.

1.2.2 Where the pollution comes from

Leather making is a wet process, and almost every operation discharges liquid. The three process groups differ so much in character that they are treated as separate streams.

Process group	Operations	Characteristic pollutants
Beamhouse	Soaking, liming, unhairing, fleshing, deliming, bating	SULPHIDE, lime, very high pH (11–12), high BOD/COD, suspended solids, hair and protein, ammonia (from deliming)
Tanyard	Pickling, chrome tanning	CHROMIUM, low pH (2.5–4), chloride, sulphate, high TDS
Post-tanning and finishing	Neutralisation, retanning, dyeing, fatliquoring, finishing	Syntans, dyes, fats and oils, residual chrome, colour

Table 1.2 The three tannery wastewater streams.

1.2.3 Typical characteristics of combined raw effluent

Parameter	Typical raw value	ECR 2023 limit (inland surface water)
pH	2.5 – 12 (stream-dependent)	6 – 9
BOD ₅	1,000 – 3,000 mg/L	50 mg/L
COD	3,000 – 8,000 mg/L	200 mg/L
Total suspended solids	2,000 – 4,000 mg/L	150 mg/L
Total dissolved solids	10,000 – 40,000 mg/L	2,100 mg/L
Sulphide	100 – 300 mg/L	2.0 mg/L
Total chromium	100 – 400 mg/L	2.0 mg/L
Oil and grease	200 – 1,000 mg/L	10 mg/L
Colour	Strong	150 Hazen

Table 1.3 Indicative composition of combined raw tannery effluent, compared with the Bangladesh Environmental Conservation Rules (ECR) 2023 discharge standards. Values vary widely with the process and the tannery.

Reading Table 1.3 tells you the size of the task. Suspended solids must fall by a factor of about twenty, BOD by forty, and sulphide and chromium by a hundred or more. Only the dissolved solids resist every stage of conventional treatment — a point returned to in Chapter 2.

TWO HAZARDS TO REMEMBER

SULPHIDE. In alkaline conditions sulphide stays safely dissolved. If the pH falls below about 8 it converts to hydrogen sulphide (H₂S), which escapes as a highly toxic gas. Mixing an acidic stream with a sulphide-bearing stream can therefore kill people in a drain or tank. This single fact governs the design of the collection system.

CHROMIUM. Trivalent chromium, Cr(III), is the form used in tanning; it is relatively insoluble and of low toxicity. Hexavalent chromium, Cr(VI), is soluble, mobile and carcinogenic. Careless handling — particularly incineration at the wrong temperature, or prolonged alkaline storage — can convert one into the other.

1.3 The Collection System and Stream Segregation

Purpose

To convey wastewater from each process to the treatment plant while keeping incompatible streams apart until they have been made safe.

Principle

Two chemical facts make segregation compulsory rather than optional.

1. **Sulphide and acid must not meet.** The beamhouse stream is alkaline and carries sulphide; the tanyard stream is acidic. Combining them releases hydrogen sulphide gas.
2. **Chromium must be caught while it is concentrated.** Spent chrome liquor contains 2,000–4,000 mg/L of chromium and can be economically recovered. Diluted into the general flow at 100–400 mg/L it can only be precipitated into the sludge, where it becomes a hazardous waste rather than a raw material.

Process description

A properly designed tannery has at least two separate drainage networks, and often three.

Drain	Carries	Goes to
Chrome drain	Spent chrome tanning and pickling liquor	Chrome recovery unit (Section 1.12)
Beamhouse / general drain	Soak, lime, deliming and bating liquors	Sulphide oxidation, then equalisation
Post-tanning / finishing drain	Dyeing, fatliquoring and finishing washings	Equalisation (may join the general drain)

Table 1.4 Segregated collection system.

Important design considerations

- Channels should be covered or grated, with adequate gradient to prevent settlement of solids.
- The beamhouse drain must be kept ALKALINE throughout its length; any acid entry point is a hazard.
- Ventilation of covered drains is necessary to prevent accumulation of H_2S .
- Screens or catch pits are placed at the point of discharge from each drum or pit, so that solids are removed where they are most concentrated.
- Each stream should be metered, so that water use can be managed and charges apportioned fairly.

Application to tannery wastewater

This is the single most important design decision in the whole treatment scheme, and it costs almost nothing compared with the plant it protects. Where segregation is neglected — as commonly happens where many small tanneries discharge into a shared sewer — chromium cannot be recovered, sulphide cannot be safely oxidised, and the entire sludge output of the plant becomes hazardous waste.

KEY IDEA

Segregate at source. Once the streams are mixed, neither the chromium nor the sulphide can be dealt with properly, and the problem becomes far larger and far more expensive.

1.4 Objectives and Classification of Primary Treatment

1.4.1 The four objectives

No.	Objective	How it is achieved
1	Remove coarse and floating material	Screening; grit removal; oil and grease removal
2	Balance flow, load and pH	Equalisation with mixing; neutralisation
3	Remove chemical hazards	Sulphide oxidation; chromium recovery
4	Remove suspended and colloidal solids	Coagulation–flocculation; sedimentation; flotation

Table 1.5 Objectives of primary treatment and the units that meet them.

1.4.2 Physical and chemical unit operations

Primary treatment uses two of the three families of unit operation. It is worth being clear about the distinction, because it explains why the units appear in the order they do.

Physical treatment — separates material by exploiting size, density or velocity. No reagent changes the pollutant chemically — the pollutant is simply moved from the water to somewhere else. Screening, grit removal, equalisation, sedimentation and flotation are physical.

Chemical treatment — adds a reagent that changes the pollutant chemically, or converts it into a form that can then be separated physically. Neutralisation, sulphide oxidation, chromium precipitation and coagulation are chemical.

Physicochemical treatment — a term used where a chemical step is applied specifically to make a physical separation possible — coagulation followed by sedimentation being the standard example.

Biological treatment, the third family, uses living micro-organisms and belongs to Chapter 3.

1.4.3 The order of the units

The sequence is not arbitrary. Each unit protects the next, and the whole train reads as follows.

Screening → Grit removal → Grease removal → Pumping → Fine screening
→ Equalisation with sulphide oxidation → Neutralisation
→ Coagulation–flocculation → Primary sedimentation → to Chapter 2 / Chapter 3
with the chrome stream treated separately in a chrome recovery unit before joining

Coarse material is removed first because it damages pumps and blocks pipes. Grit is removed before pumping because it abrades impellers. Sulphide is oxidised before the streams are combined and before the pH is lowered. Coagulation comes late, when the flow is steady and the pH is correct, because both matter greatly to its performance. Sedimentation comes last, to capture what coagulation produced.

1.5 Screening

Purpose

To intercept large, floating and fibrous material before it reaches pumps, pipes and downstream tanks.

Principle

A screen is a physical barrier of bars or mesh which stops solids larger than its openings while allowing water to pass. The mechanism is STRAINING only; nothing is changed chemically. It is the first unit in any treatment plant.

Process description and components

Wastewater flows through an inclined bar rack or mesh set across the channel. Retained material — called SCREENINGS — accumulates on the upstream face and is removed by a rake, either manually or by a mechanical raking mechanism, and discharged to a bin for disposal as solid waste.

Type	Opening size	Removes	Cleaning
Coarse screen (bar rack)	25 – 100 mm	Fleshings, hair mats, hide trimmings, plastic	Mechanical rake or manual
Medium screen	6 – 25 mm	Smaller trimmings and debris	Mechanical
Fine screen	1.5 – 6 mm	Hair, fibre, finer solids	Mechanical, often rotating drum or step screen

Table 1.6 Classification of screens by opening size.

Important operating parameters

- **Bar spacing** — sets what is caught; too fine a spacing on a coarse screen blinds rapidly.
- **Approach velocity** — typically 0.3–0.6 m/s. Too low and grit settles in the channel; too high and material is forced through the bars.
- **Velocity through the openings** — should not exceed about 0.9 m/s.
- **Head loss** — the drop in water level across the screen, which rises as the screen blinds.

Head loss through a bar screen is estimated by:

$$h_f = (1/0.7) \cdot (V^2 - v^2) / 2g$$

V = velocity through the openings (m/s); v = approach velocity (m/s); g = 9.81 m/s²; 0.7 is a discharge coefficient

The practical lesson from this equation is that head loss depends on the SQUARE of the velocity. As the screen blinds with debris, the open area falls, V rises, and the head loss rises very sharply. Regular cleaning is therefore not a matter of tidiness but of hydraulic capacity.

Pollutants removed

Gross suspended solids — typically 30–40% of the total suspended solids in a tannery effluent, because so much of that load is coarse hair, flesh and trimmings.

Advantages and limitations

Advantages	Limitations
Simple, robust, low capital cost	Requires regular cleaning; blinds if neglected
Protects all downstream equipment	Screenings are a solid waste needing disposal
No chemicals or energy (gravity screens)	Removes only coarse matter — no dissolved load
Immediate and visible benefit	Odour and fly nuisance if screenings are stored

Application to tannery wastewater

Screening matters more in a tannery than in most industries because the beamhouse discharges very large quantities of hair, flesh and lime sludge. Screens are best placed at each drum or pit outlet as well as at the plant inlet, since solids are easiest to remove where they are most concentrated. Screenings are dewatered and disposed of as solid waste; where hair-save unhairing is practised, the recovered hair may itself be a saleable by-product.

1.6 Grit Removal

Purpose

To remove heavy, dense inorganic particles — sand, silt, lime grit and dirt washed off hides — which would otherwise abrade pumps and accumulate in tanks.

Principle

Grit is far denser than organic solids: about 2,650 kg/m³ against roughly 1,050 for organic matter. It therefore settles much faster. A grit chamber exploits this by slowing the flow to a velocity at which the dense grit settles but the lighter organic matter remains in suspension and passes on.

Process description

The wastewater passes through a channel or tank in which the horizontal velocity is controlled at about 0.3 m/s. Grit falls to the floor and is removed by scraper, screw conveyor or airlift. Three arrangements are common:

- **Horizontal-flow grit chamber** — a long channel with velocity control by a proportional weir.
- **Aerated grit chamber** — air is introduced along one side to create a spiral flow; grit falls out while organics stay suspended. Also strips some grease.
- **Vortex grit chamber** — flow enters tangentially, and centrifugal action concentrates grit at the centre. Compact.

Important operating parameters

Parameter	Typical value	Why it matters
Horizontal velocity	≈ 0.3 m/s	Too low: organics settle too. Too high: grit passes through
Detention time	45 – 90 s (horizontal)	Sufficient for grit to reach the floor
Grit particle size removed	≥ 0.2 mm	Design basis
Air supply (aerated type)	0.2 – 0.5 m ³ /min per m of length	Sets the spiral velocity

Advantages and limitations

Grit removal is inexpensive and greatly extends the life of pumps and mechanical equipment. Its limitations are that the velocity must be controlled within a fairly narrow band, and that in a tannery the removed grit is contaminated with organic matter and lime and must be washed or disposed of carefully.

Application to tannery wastewater

Tannery grit consists largely of dirt and sand carried in on raw hides, together with undissolved lime particles. It is placed after coarse screening and before pumping, so that pumps handle water free of abrasive material.

1.7 Oil and Grease Removal

Purpose

To remove fats, oils and grease (FOG) which float on the surface, coat equipment, block pipes and interfere with both settling and biological treatment.

Principle

Oil and grease are less dense than water, so under quiescent conditions they rise to the surface and can be skimmed off. Where the oil is emulsified — dispersed as fine droplets that will not rise on their own — flotation must be assisted, either chemically (by breaking the emulsion) or physically (by attaching air bubbles, as in Section 1.15).

Process description

A grease trap is a **quiescent** chamber, often with baffles arranged so that water leaves from below the surface while the floating layer is retained. The accumulated grease is skimmed manually or by a surface scraper. In a larger plant an aerated grit chamber or a dissolved-air flotation unit performs the same duty more effectively.

Important operating parameters

- Detention time: 15–30 minutes for a simple grease trap.
- Surface loading: low enough for droplets to rise — typically under $20 \text{ m}^3/\text{m}^2 \cdot \text{h}$.
- Frequency of skimming: accumulated grease must be removed regularly or it is re-entrained.

Application to tannery wastewater

The fat content of tannery effluent comes chiefly from fleshing and from fatliquoring. Fleshing liquors in particular carry a very heavy grease load. Removing it early prevents grease from coating the walls of equalisation tanks, from interfering with floc formation in coagulation, and from causing the foaming problems described in Chapter 3. Recovered fat has some value as a raw material for tallow or biodiesel.

1.8 Flow Equalisation and Flow Regulation

Purpose

To convert a highly variable discharge into a steady, uniform flow of uniform composition, so that every downstream unit receives a constant load.

Principle

Tannery production is a BATCH process. A drum is charged, run, and dumped; each dump releases a large volume of concentrated liquor in a short time, and each has a different composition. Flow may vary from zero to several times the average within an hour, and the pH may swing from 12 to 3 between one dump and the next.

An equalisation tank absorbs these swings. It stores the incoming batches, mixes them, and releases the blended liquor at a controlled, constant rate. The tank acts as a hydraulic and chemical BUFFER — the same principle as a shock absorber.

KEY IDEA

Equalisation is the most important single unit in a tannery primary treatment plant. Chemical dosing, coagulation, settling and above all biological treatment are all designed for a particular flow and load; if the flow and load vary wildly, none of them can be controlled. Without equalisation, every later unit is being asked to hit a moving target.

Process description and components

A large tank, usually rectangular and 3–5 m deep, receives the combined (or per-stream) flow. It is continuously mixed, and the outlet flow is controlled by a pump or a fixed-level weir. Three functions occur simultaneously:

- **Flow balancing** — peaks are stored and troughs are made up, giving a constant outflow.
- **Load homogenisation** — concentrated and dilute batches are blended, so BOD, COD and pH are averaged.
- **Sulphide oxidation** — where air is used for mixing, the tank simultaneously serves as the sulphide oxidation reactor (Section 1.11).

Important operating parameters

Parameter	Typical value	Comment
Volume	≈ one day's total discharge	Plus 20% freeboard
Depth	3 – 5 m	Deeper needs more mixing energy
Mixing power	4 – 8 W/m ³ (mechanical)	Enough to keep solids in suspension
Air supply (diffused)	0.6 – 1.2 m ³ air per m ³ per hour	Also oxidises sulphide
Retention time	8 – 24 h	Longer for greater variability

Sizing an equalisation tank

The required volume is found from the cumulative inflow and outflow over a full production cycle. In practice, for a tannery, the working rule is that the tank should hold approximately one day's discharge, with freeboard added.

WORKED EXAMPLE 1.1 — Equalisation tank

A tannery discharges 200 m³/d. Size the equalisation tank.

Step 1 Working volume = one day's flow = 200 m³

Step 2 Add 20% freeboard: $200 \times 1.20 = 240 \text{ m}^3$

Step 3 At a working depth of 4 m, plan area = $240 / 4 = 60 \text{ m}^2$

Step 4 Adopt a tank approximately 7.7 m × 7.8 m × 4 m deep

Step 5 Mixing power at 6 W/m³: $240 \times 6 = 1,440 \text{ W} \approx 1.5 \text{ kW}$ installed

ANSWER: an equalisation tank of about 240 m³, 7.7 × 7.8 × 4 m, with roughly 1.5 kW of mixing.

Advantages and limitations

Advantages	Limitations
Protects every downstream unit from shock	Large tank — significant land and capital
Allows accurate chemical dosing	Continuous mixing energy required
Dilutes toxic slugs before they reach the biology	Odour if mixing fails and the tank goes septic
Provides the reactor for sulphide oxidation	Solids settle and accumulate if under-mixed

Application to tannery wastewater

Because tannery discharge is exceptionally batchy and chemically extreme, equalisation is indispensable. It is generally the largest tank in the primary section. Where the beamhouse stream is equalised separately, the tank doubles as the sulphide oxidation reactor; where streams are equalised together, sulphide must be oxidised first.

1.9 Mixing

Purpose

To apply energy so that a tank is uniform — so that an added chemical reaches every droplet, or a variable stream is blended smooth.

Principle

Mixing is not a treatment process in itself but a supporting operation on which several others depend. Two regimes are distinguished, and confusing them is a common source of poor performance.

Regime	Character	Purpose
Rapid (flash) mixing	Violent, brief — seconds	Disperse a coagulant instantly before it is consumed locally
Slow mixing (flocculation)	Gentle, prolonged — minutes	Grow floc by promoting collisions without tearing it apart
Maintenance mixing	Moderate, continuous	Keep solids suspended in an equalisation tank

Quantifying mixing: the velocity gradient

Mixing intensity is measured by the VELOCITY GRADIENT G , the rate at which adjacent layers of fluid shear past one another:

$$G = \sqrt{P / (\mu V)}$$

G = velocity gradient (s^{-1}); P = power input (W); μ = dynamic viscosity ($Pa \cdot s$); V = tank volume (m^3)

Application	G (s^{-1})	Contact time
Rapid mixing (coagulation)	300 – 1,000	20 – 60 s
Flocculation	20 – 80	15 – 30 min
Equalisation tank mixing	≈ 50 – 100	continuous

Table 1.7 Typical velocity gradients.

The product $G \cdot t$ (velocity gradient multiplied by time) is often used as a design criterion for flocculation, with values in the range 10,000–100,000 being typical.

Equipment

- Mechanical: paddle, turbine or propeller mixers driven by a motor.
- Hydraulic: baffled channels, weirs and hydraulic jumps that generate turbulence without moving parts.
- Pneumatic: diffused air, which mixes and simultaneously supplies oxygen.

1.10 Neutralisation and pH Correction

Purpose

To bring the wastewater to a pH acceptable to the downstream processes and to the discharge standard.

Principle

Neutralisation is the reaction of an acid with a base to produce a salt and water. Where the wastewater is alkaline, acid is added; where it is acidic, alkali is added. The target range is set by what comes next:

- Biological treatment requires pH 6.5–8.5 (micro-organisms are inhibited outside this).
- Coagulation with alum works best at pH 6–7.5; with ferric chloride, pH 5–9.
- The ECR 2023 discharge limit is pH 6–9.

Process description and reagents

Direction	Reagent	Notes
To raise pH	Lime, CaO or Ca(OH) ₂	Cheapest; also precipitates metals; produces more sludge
To raise pH	Sodium hydroxide, NaOH	Fast and clean; more costly; no added sludge
To lower pH	Sulphuric acid, H ₂ SO ₄	Common and cheap; adds sulphate
To lower pH	Carbon dioxide (flue gas)	Mild, self-limiting, cannot overshoot

Lime: a special note

Lime deserves separate mention because it does several jobs at once in a tannery plant. Added as quicklime (CaO) it slakes to calcium hydroxide:



Calcium hydroxide then raises the pH, precipitates metal hydroxides, assists coagulation, and helps remove phosphate and some sulphate. Its drawback is that it generates a substantial volume of additional sludge, and it is difficult to handle as a slurry.

Important operating parameters

- Target pH and permitted tolerance.
- Reagent dose — always determined by titration on the actual wastewater, never assumed, because tannery effluent is strongly buffered.
- Mixing: rapid mixing is essential so that the reagent is dispersed before it reacts locally.
- Control: pH is measured continuously and the dosing pump adjusted automatically; manual dosing invariably overshoots.

A PRACTICAL WARNING

Neutralisation of a sulphide-bearing stream must NEVER be carried out before the sulphide has been oxidised. Lowering the pH of a sulphide liquor releases hydrogen sulphide gas. Sulphide oxidation (Section 1.11) always precedes acid dosing.

Application to tannery wastewater

Because the beamhouse stream is strongly alkaline and the tanyard stream strongly acidic, a tannery has an unusual opportunity: the two streams can be used to neutralise one another. Controlled blending in the equalisation tank — AFTER sulphide oxidation — substantially reduces the quantity of reagent that must be purchased. This is one of the clearest economies available in tannery effluent treatment.

1.11 Sulphide Oxidation

Purpose

To convert toxic sulphide into harmless sulphate before the beamhouse stream is mixed with other streams or has its pH lowered.

Principle

Sulphide is oxidised by dissolved oxygen to sulphate:



The reaction is slow in plain water but is greatly accelerated by a catalyst, usually a manganese salt (manganese sulphate) at 50–100 mg/L. The process is therefore known as CATALYTIC AIR OXIDATION.

Note the stoichiometry: two moles of oxygen (64 g) per mole of sulphide (32 g), that is about 2 kg of oxygen per kg of sulphide. The process is air-intensive rather than chemical-intensive, which is what makes it economical.

Process description

The sulphide-bearing liquor is held in a tank, kept strongly alkaline (pH above 9, to prevent any H₂S release), dosed with the manganese catalyst, and vigorously aerated. Air may be supplied through fine-bubble diffusers or, more commonly in tanneries, by a venturi or jet aerator which entrains air into a recirculating stream and provides intense contact.

Important operating parameters

Parameter	Typical value	Comment
pH	> 9 throughout	Below 8, H ₂ S is released
Catalyst (MnSO ₄)	50 – 100 mg/L	Manganese; recyclable with the liquor
Oxygen requirement	≈ 2 kg O ₂ per kg sulphide	From stoichiometry
Reaction time	6 – 8 h (batch)	Until sulphide < 1 mg/L
Final sulphide	< 1 – 2 mg/L	ECR 2023 limit is 2.0 mg/L

Alternative methods

Method	Principle	Assessment
Catalytic air oxidation	Air + Mn catalyst	Preferred: cheap, no added sludge
Ferrous sulphate precipitation	Fe ²⁺ + S ²⁻ → FeS↓	Effective but produces large volumes of sludge
Hydrogen peroxide	Chemical oxidation	Fast and clean but expensive
Hair-save unhairing	Avoids dissolving the hair	Reduces sulphide at source by 30–50% — the best option of all

Advantages and limitations

Catalytic air oxidation is inexpensive to run, adds almost no sludge, and can be carried out in the equalisation tank itself. Its limitations are the air demand, the batch reaction time, and the need to maintain a high pH throughout — an operational discipline that is easily neglected.

Application to tannery wastewater

This unit is indispensable and non-negotiable. Sulphide is toxic to the micro-organisms of Chapter 3, exerts an oxygen demand, corrodes concrete and metalwork, and generates a gas that has killed workers in tannery drains. Oxidation must be completed BEFORE the beamhouse stream meets any acidic stream and before neutralisation.

1.12 Chromium Recovery and Precipitation

Purpose

To recover chromium from the segregated spent chrome liquor, both to reclaim a valuable tanning chemical and to keep chromium out of the general effluent and the mixed sludge.

Principle

Chromium in spent tanning liquor is already in the trivalent state, Cr(III). It is soluble at the low pH of the liquor (about 3.5–4.0). Raising the pH converts it to an insoluble hydroxide which can be settled and separated:



The precipitate is then redissolved in sulphuric acid to regenerate a usable chrome tanning liquor.

A caution about terminology: the word ‘reduction’ in this context means reduction of the chromium LOAD in the effluent, not a change of oxidation state. No redox reaction occurs — the chromium is trivalent before and after.

Process description

1. Collect the spent chrome liquor separately, through the dedicated chrome drain.
2. Screen it to remove hide fibre and other solids.
3. BASIFICATION: add alkali slowly with stirring, raising the pH to 8.0–8.5. Magnesium oxide (MgO) is preferred to NaOH because it raises the pH gently and produces a denser, faster-settling floc.
4. Allow the $\text{Cr}(\text{OH})_3$ precipitate to settle (1–2 hours); decant the supernatant.
5. Dewater the settled chrome cake.
6. REDISSOLUTION: dissolve the cake in concentrated sulphuric acid to a pH of about 2.5, regenerating basic chromium sulphate liquor.
7. Analyse the regenerated liquor for chromium content, and blend it with fresh chrome for reuse in tanning.

Important operating parameters

Parameter	Typical value	Comment
Basification pH	8.0 – 8.5	Below 7, precipitation is incomplete
Maximum pH	< 9.5	Above this $\text{Cr}(\text{OH})_3$ redissolves as chromite
Alkali	MgO preferred; NaOH acceptable	MgO gives denser sludge

Parameter	Typical value	Comment
Settling time	1 – 2 h	The floc is gelatinous and slow
Recovery efficiency	95 – 98%	Of the chromium in the treated liquor
Redissolution acid	H ₂ SO ₄ to pH ≈ 2.5	Restores tanning liquor

KEY IDEA

There is an optimum pH window. Below about pH 7 the chromium does not fully precipitate; above about pH 9.5 the precipitate begins to redissolve as soluble chromite. Overshooting the dose therefore **LOWERS** the recovery. This is why the alkali is added slowly, with the pH monitored continuously.

Advantages and limitations

Advantages	Limitations
Recovers 95–98% of a costly chemical	Requires strict stream segregation
Removes the hazardous fraction from the general sludge	Impurities accumulate over repeated cycles
Reduces hazardous-waste disposal cost	Needs careful pH control and trained operators
Supports LWG and buyer compliance requirements	Regenerated liquor usually needs blending with fresh chrome

Application to tannery wastewater

Chrome recovery is the clearest example in the whole of effluent treatment where an environmental measure pays for itself. Its feasibility depends entirely on the segregation described in Section 1.3. Where it is practised, the general sludge is non-hazardous and may be landfilled normally; where it is not, every tonne of sludge from the whole plant is classified as hazardous waste.

1.13 Coagulation and Flocculation

Purpose

To convert fine colloidal particles — which will never settle on their own — into larger aggregates that can be removed by sedimentation or flotation.

1.13.1 The problem: why colloids do not settle

Colloid — a particle roughly 1 nm to 1 µm across, small enough that gravity has almost no effect on it. Colloids cause the turbidity and much of the colour of an effluent.

Colloids remain suspended indefinitely for a definite physical reason. Each particle carries a surface electrical charge — in wastewater, almost always negative. Like charges repel, so the particles push one another apart and never collide long enough to join. The surrounding cloud of counter-ions is called the ELECTRICAL DOUBLE LAYER, and its net charge is the ZETA POTENTIAL. A high zeta potential means a stable, well-dispersed suspension that will not settle.

1.13.2 Coagulation

Coagulation — the chemical destabilisation of colloids by adding a coagulant that neutralises their surface charge, so that the repulsion disappears and particles can approach one another.

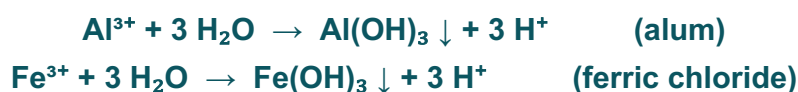
Coagulants are salts of trivalent metals, which supply highly charged ions. Four mechanisms act together:

- **Charge neutralisation** — the positive metal ions neutralise the negative colloid surface.
- **Double-layer compression** — added ions compress the ionic cloud, reducing the repulsive range.
- **Sweep flocculation** — the growing metal-hydroxide precipitate physically enmeshes colloids as it settles, like a net drawn through the water.
- **Interparticle bridging** — long-chain polymers attach to several particles at once and bind them together.

Coagulation requires RAPID MIXING, because the coagulant hydrolyses within seconds and must be dispersed through the whole volume before it is consumed locally.

1.13.3 The coagulants

Two are used in tannery practice, and their hydrolysis reactions explain their behaviour.



Both release hydrogen ions, so both LOWER the pH and consume alkalinity. This must be allowed for in the neutralisation stage.

Property	Potash alum, $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$	Ferric chloride, FeCl_3
Effective ion	Al^{3+}	Fe^{3+}
Working pH range	6.0 – 7.5 (narrow)	5 – 9 (wide)
Floc character	Lighter, slower settling	Denser, faster settling
Colour removal	Moderate	Good
Typical dose	200 – 600 mg/L	200 – 600 mg/L
Drawback	Narrow pH window	May impart a residual colour; corrosive

1.13.4 Flocculation

Flocculation — the slow, gentle mixing that follows coagulation, during which destabilised particles collide and grow into large, settleable aggregates called FLOCS.

The distinction from coagulation is important and often confused. Coagulation is CHEMICAL destabilisation requiring fast mixing; flocculation is PHYSICAL aggregation requiring slow mixing. Too vigorous a stir during flocculation tears the growing flocs apart.

A FLOCCULANT or coagulant aid — usually a long-chain polyelectrolyte dosed at 0.5–2 mg/L — may be added to build larger and stronger flocs by bridging.

1.13.5 Determining the dose: the jar test

The correct dose cannot be calculated from theory; it must be determined experimentally on the actual wastewater by the JAR TEST, which reproduces the full-scale process at bench scale (reference standard ASTM D2035).

1. Fill several beakers with equal volumes of the wastewater (typically 500 mL or 1 L).
2. Dose each beaker with a different quantity of coagulant, prepared as a solution (never as powder — powder cannot be dosed accurately or dispersed quickly).
3. Rapid mix, about 120 rpm for 1 minute.
4. Slow mix, about 30 rpm for 20 minutes.
5. Allow to settle undisturbed for 30 minutes.
6. Draw supernatant from a fixed depth and measure turbidity, colour or COD.
7. Plot removal against dose. The OPTIMUM DOSE is the lowest dose giving the required removal — the point at which the curve levels off.

WORKED EXAMPLE 1.2 — Preparing and dosing coagulant

A jar test is to be run at doses of 200, 400 and 600 mg/L in 500 mL beakers. How much stock solution is added to each?

Step 1 Prepare a stock solution of 10 g/L: dissolve 10 g of coagulant and make up to 1 litre.
Therefore 1 mL of stock contains 10 mg of coagulant.

Step 2 Volume of stock required = dose (mg/L) × beaker volume (L) ÷ 10

$$200 \text{ mg/L} \times 0.5 \text{ L} = 100 \text{ mg} \rightarrow 100 / 10 = 10 \text{ mL}$$

$$400 \text{ mg/L} \times 0.5 \text{ L} = 200 \text{ mg} \rightarrow 200 / 10 = 20 \text{ mL}$$

$$600 \text{ mg/L} \times 0.5 \text{ L} = 300 \text{ mg} \rightarrow 300 / 10 = 30 \text{ mL}$$

ANSWER: add 10, 20 and 30 mL of the 10 g/L stock respectively.

Note: if the added volume matters, take (500 – V) mL of wastewater so that every beaker finishes at the same total volume.

1.13.6 Important operating parameters

Parameter	Typical value	Effect if wrong
Coagulant dose	200 – 600 mg/L (jar-test determined)	Underdose: incomplete. Overdose: charge reversal and re-dispersion
pH	Alum 6–7.5; ferric 5–9	Outside range, the hydroxide does not form properly
Rapid mix	G 300–1,000 s ⁻¹ , 20–60 s	Poor dispersion; coagulant wasted
Slow mix	G 20–80 s ⁻¹ , 15–30 min	Too fast tears floc; too slow gives no collisions
Polyelectrolyte	0.5 – 2 mg/L	Overdose re-stabilises the suspension

1.13.7 Advantages, limitations and tannery application

Advantages	Limitations
Removes colloids that no physical process can	Chemical cost is continuous
Removes 50–70% of COD and much colour	Generates a large volume of chemical sludge
Also precipitates residual chromium	Dose must be re-determined as the effluent changes
Improves settling dramatically	Lowers pH; alkalinity must be replaced

In a tannery, coagulation–flocculation is the principal chemical step of primary treatment. It removes the fine solids and part of the colour and COD that screening and settling cannot, and it precipitates residual chromium that escaped the recovery unit. Its cost — in chemicals and in sludge — is substantial, which is a further argument for reducing the load at source.

1.14 Sedimentation and Primary Clarification

Purpose

To separate suspended and flocculated solids from the water by gravity, producing a clarified effluent and a concentrated sludge.

1.14.1 Principle

A particle denser than water sinks when the downward gravitational force exceeds the upward drag of the surrounding liquid. For a discrete spherical particle in laminar flow, the terminal settling velocity is given by Stokes' law:

$$v = g (\rho_p - \rho_v) d^2 / 18 \mu$$

v = settling velocity (m/s); ρ_p , ρ_v = densities of particle and water (kg/m^3); d = particle diameter (m); μ = dynamic viscosity ($\text{Pa}\cdot\text{s}$)

Three consequences follow directly, and they explain most of what happens in a clarifier:

- **Velocity varies with the SQUARE of diameter.** Doubling the particle size settles it four times faster. This is precisely why coagulation — which makes small colloids into large flocs — transforms settling performance.
- **Denser particles settle faster.** Grit settles in seconds; organic floc takes an hour.
- **Warmer water settles faster.** Viscosity falls as temperature rises, so performance is better in summer.

1.14.2 Types of settling

Type	Description	Where it occurs
Type I — discrete	Particles settle independently at constant velocity; no interaction	Grit chambers; dilute suspensions
Type II — flocculent	Particles collide and coalesce as they fall, settling progressively faster	Primary clarifiers; coagulated water
Type III — hindered (zone)	Particles settle as a mass with a distinct interface	Thickeners; secondary clarifiers
Type IV — compression	Particles rest on one another; the layer consolidates under its own weight	Bottom of thickeners

Table 1.9 The four types of settling.

Primary sedimentation of tannery effluent is chiefly Type II, which is why the settling curve is not a straight line: the flocs grow as they fall and accelerate.

1.14.3 Process description and components

A primary clarifier is a quiescent tank, most commonly circular, in which the flow is slowed enough for solids to fall to the floor while the clarified liquid overflows at the surface. The main components are:

- **Inlet and centre feed well** — distributes the flow evenly and dissipates its energy so that it does not disturb the settled sludge.
- **Settling zone** — the main quiescent volume where separation occurs.
- **Sludge zone and hopper** — a sloping floor leading to a central hopper from which sludge is withdrawn.
- **Scraper mechanism** — rotates slowly, moving settled sludge to the hopper and skimming any floating material.

- **Overflow weirs and launder** — collect the clarified water uniformly around the perimeter.

1.14.4 Important operating parameters

The governing design parameter is the SURFACE OVERFLOW RATE (also called surface loading rate), which is the flow divided by the surface area:

$$\text{SOR} = Q / A \quad (\text{m}^3/\text{m}^2 \cdot \text{d}, \text{ equivalent to m/d})$$

The physical meaning repays attention: the surface overflow rate is the upward velocity of the rising water. Any particle whose settling velocity exceeds the SOR will be captured; any slower particle will be carried out. It follows that clarifier performance depends on SURFACE AREA, not on depth — a deeper tank of the same area removes no more solids, though depth does provide sludge storage and buffer against disturbance.

Parameter	Typical value	Comment
Surface overflow rate	20 – 40 m ³ /m ² ·d	The governing parameter
Detention time	1.5 – 2.5 h	Time for flocs to reach the floor
Depth (side water)	3 – 4 m	Provides sludge storage
Weir loading rate	< 125 – 250 m ³ /m·d	Too high draws solids over the weir
Scraper tip speed	0.02 – 0.05 m/s	Slow, to avoid re-suspension
Sludge withdrawal	Continuous or intermittent	Sludge must not be left to go septic

WORKED EXAMPLE 1.3 — Sizing a primary clarifier

Size a circular primary clarifier for a tannery discharging 400 m³/d.

Step 1 Adopt a surface overflow rate of 25 m³/m²·d

Step 2 Required area $A = Q / \text{SOR} = 400 / 25 = 16 \text{ m}^2$

Step 3 Diameter $D = \sqrt{(4A / \pi)} = \sqrt{(4 \times 16 / 3.1416)} = \sqrt{20.4} = 4.5 \text{ m}$

Adopt 4.5 m diameter ($A = 15.9 \text{ m}^2$)

Step 4 Check detention time at 3.5 m depth:

Volume = $15.9 \times 3.5 = 55.7 \text{ m}^3$

$t = 55.7 / 400 = 0.139 \text{ d} = 3.3 \text{ hours}$ — comfortably adequate

Step 5 Check weir loading, weir length = $\pi D = 14.1 \text{ m}$:

$400 / 14.1 = 28 \text{ m}^3/\text{m} \cdot \text{d}$ — well within the limit

ANSWER: a circular clarifier 4.5 m in diameter and 3.5 m deep.

1.14.5 Performance

Condition	TSS removal	BOD removal
Plain sedimentation (no chemicals)	50 – 65%	25 – 35%
With coagulation–flocculation	70 – 90%	40 – 60%

Table 1.10 Typical primary sedimentation performance.

The contrast between the two rows is the quantitative justification for Section 1.13: chemical assistance roughly doubles the useful work a clarifier does.

1.14.6 Advantages, limitations and application

Advantages	Limitations
Simple, reliable, low energy (gravity only)	Large surface area required
Removes the bulk of settleable solids	Removes no dissolved matter
Thickens sludge as well as clarifying water	Sludge goes septic if not withdrawn regularly
Buffers the downstream units against solids surges	Colloids pass through unless coagulated

In a tannery scheme the primary clarifier follows coagulation–flocculation, and its function is to capture the chemical floc together with the residual settleable solids. It is the last unit of primary treatment; what leaves it is a clarified but still heavily polluted water, carrying the dissolved and colloidal load which Chapters 2 and 3 must address.

1.15 Flotation (Dissolved Air Flotation)

Purpose

To remove particles and oil droplets that are too light to settle, by carrying them UPWARDS instead of downwards.

Principle

Sedimentation fails for particles whose density is close to that of water — fine grease, emulsified oil, light fibre. Flotation reverses the problem. Fine air bubbles are introduced which attach to the particles, so that the combined bubble-particle aggregate becomes lighter than water and rises to the surface, where it is skimmed off as a scum.

In DISSOLVED AIR FLOTATION (DAF), air is dissolved into a stream of water under pressure (typically 4–6 bar) and this stream is then released into the flotation tank at atmospheric pressure. The sudden pressure drop causes the dissolved air to come out of solution as a cloud of very fine bubbles, 30–50 μm across — small enough to attach to fine particles.

Process description and components

- Saturator (pressure vessel) in which air is dissolved into a recycle stream.
- Pressure release valve or nozzle, where the bubbles are formed.
- Flotation tank, in which the bubbles rise and carry the solids with them.
- Surface skimmer, which removes the accumulated float (scum).
- Bottom sludge draw-off, for the fraction that still settles.

Important operating parameters

Parameter	Typical value	Comment
Saturation pressure	4 – 6 bar	Governs how much air dissolves
Recycle ratio	10 – 30% of feed	The pressurised stream
Air-to-solids ratio	0.02 – 0.06 kg air/kg solids	The key design parameter
Surface loading	5 – 10 $\text{m}^3/\text{m}^2\cdot\text{h}$	Higher than a clarifier
Bubble size	30 – 50 μm	Fine bubbles attach; coarse ones do not

Advantages, limitations and application

Flotation is much faster than sedimentation and needs far less area, and it removes grease and light solids that gravity cannot. Against this it consumes energy for the saturator and compressor, is mechanically more complex, and requires skilled operation.

In tannery practice DAF is used chiefly where the effluent is rich in fats and oils — for example on fleshing and fatliquoring streams — and in footwear factories, where adhesives, oils and finishing chemicals make oil and grease the signature pollutant. It is often used in combination with coagulation, the chemical floc being floated rather than settled.

1.16 Primary Sludge: Quantities and Handling

The principle to remember

Primary treatment does not destroy pollution. It CONCENTRATES it: material that was dispersed through a large volume of water is collected into a small volume of sludge. Everything removed from the water in Sections 1.5 to 1.15 is still on the site, and must still be dealt with.

Sources and character

Source	Nature	Dry solids	Note
Screenings	Hair, flesh, trimmings	—	Handled as solid waste
Grit	Sand, silt, lime	—	Washed and landfilled
Grease skimmings	Fats and oils	—	May be recovered as tallow
Chrome cake	Cr(OH) ₃	5 – 15%	Recovered for reuse
Primary sludge	Settled solids + chemical floc	2 – 6%	The main stream

Quantity

Primary sludge production may be estimated from the solids removed:

$$P_s = Q \times (TSS_{in} - TSS_{out}) \times \text{removal efficiency}$$

A tannery treating 400 m³/d with an influent TSS of 3,000 mg/L and 70% removal produces:

$$400 \times 3.0 \text{ kg/m}^3 \times 0.70 = 840 \text{ kg of dry solids per day}$$

At 4% solids this is 21 m³/d of wet sludge — which explains why dewatering, not treatment, is often the largest operating cost of a small plant. Chemical coagulation adds substantially to this figure.

Handling

The sequence is thickening, then conditioning, then dewatering (by filter press, belt press or drying beds), then disposal. Detailed treatment of sludge processing belongs to the tertiary and sludge-management stage and is not repeated here. Two points specific to primary sludge should be noted:

- Primary sludge is readily dewatered compared with biological sludge, because it contains coarser, denser material.
- If chromium has NOT been segregated and recovered, the primary sludge is chrome-bearing and must be managed as hazardous waste — which multiplies its disposal cost.

1.17 Comparison of Primary Unit Operations

Unit	Type	Removes	Mechanism	Typical removal
Screening	Physical	Coarse solids, hair, fibre	Straining	30–40% of TSS
Grit removal	Physical	Sand, silt, lime grit	Differential settling	Most grit > 0.2 mm
Grease removal	Physical	Fats, oils, grease	Flotation, skimming	50–80% of FOG
Equalisation	Physical	— (balances flow and load)	Storage and mixing	—
Neutralisation	Chemical	— (corrects pH)	Acid–base reaction	pH to 6–9
Sulphide oxidation	Chemical	Sulphide	Catalytic air oxidation	To < 1–2 mg/L
Chrome recovery	Chemical	Chromium	Hydroxide precipitation	95–98%
Coagulation–floc.	Physicochemical	Colloids, colour, some COD	Charge neutralisation	50–70% of COD
Sedimentation	Physical	Suspended solids, floc	Gravity settling	50–90% of TSS
Flotation (DAF)	Physical	Oil, light solids	Bubble attachment	80–95% of FOG

Table 1.11 Summary comparison of primary treatment unit operations.

1.18 Chapter Summary

What primary treatment achieves

Parameter	Raw effluent	After primary treatment
pH	2.5 – 12	6 – 9 (corrected)
Suspended solids	2,000 – 4,000 mg/L	400 – 800 mg/L
BOD ₅	1,000 – 3,000 mg/L	800 – 1,200 mg/L
COD	3,000 – 8,000 mg/L	2,500 – 4,000 mg/L
Sulphide	100 – 300 mg/L	< 1 – 2 mg/L
Chromium	100 – 400 mg/L	Largely recovered
Dissolved solids	10,000 – 40,000 mg/L	Unchanged

Table 1.12 Typical condition of tannery effluent before and after primary treatment.

Two rows of that table define the work remaining. The BOD and COD are still far above the discharge limits, because most of the remaining organic matter is DISSOLVED and no physical process can remove it — this is the task of biological treatment, Chapter 3. And the dissolved solids are entirely unchanged, because nothing in primary treatment removes salt — a problem addressed in Chapter 2.

The main points of the chapter

1. Primary treatment removes coarse, floating and settleable material, balances the flow, corrects the pH and destroys the chemical hazards. It does not produce dischargeable water.
2. Tannery streams must be SEGREGATED at source. Sulphide and acid must never meet; chromium must be caught while concentrated.
3. Screening, grit removal and grease removal protect the plant physically.
4. Equalisation is the most important unit: it converts a batchy, chemically extreme discharge into a steady feed that every later unit can be designed for.
5. Sulphide is oxidised catalytically to sulphate BEFORE mixing or acidification.
6. Chromium is precipitated as $\text{Cr}(\text{OH})_3$ at pH 8.0–8.5 and redissolved in acid for reuse — recovering 95–98% and keeping the sludge non-hazardous.
7. Coagulation destabilises colloids chemically (fast mixing); flocculation grows them physically (slow mixing). The dose is found by jar test.
8. Sedimentation is governed by surface overflow rate, not by depth, and follows Stokes' law — which is why coagulation improves it so dramatically.
9. All of this concentrates the pollution into sludge, which must then be dewatered and disposed of responsibly.

THE CENTRAL LESSON OF CHAPTER 1

The success of the entire treatment plant is decided in primary treatment. Sulphide oxidation, chrome recovery, stream segregation and equalisation are what keep the biological stage alive and what determine whether the plant's sludge is a recoverable raw material or a hazardous waste.

And the cheapest pollution to treat is always the pollution never created — hair-save unhairing, high-exhaustion chrome tanning and reduced water use lower every load described in this chapter.

Review questions

1. Why must tannery wastewater streams be segregated at source? Give two chemical reasons.
2. Explain, using the head-loss equation, why a blinded screen loses capacity so rapidly.
3. A grit chamber operates at 0.3 m/s. What happens if the velocity falls to 0.1 m/s, and if it rises to 0.6 m/s?
4. Why is equalisation described as the most important unit in tannery primary treatment?
5. Size an equalisation tank for a tannery discharging 350 m³/d at 4.5 m depth with 20% freeboard.
6. Write the sulphide oxidation reaction and calculate the oxygen required to oxidise 40 kg of sulphide per day.
7. Explain why chromium recovery efficiency falls if the basification pH exceeds 9.5.
8. Distinguish coagulation from flocculation in terms of mechanism and mixing requirement.
9. A jar test uses 1 L beakers and a 10 g/L stock. What volumes give doses of 150, 300 and 450 mg/L?
10. State Stokes' law and explain why coagulation improves sedimentation so greatly.
11. Size a circular clarifier for 600 m³/d at a surface overflow rate of 30 m³/m²·d, and check the detention time at 3.5 m depth.
12. Why does a deeper clarifier of the same area not remove more solids?
13. When would you choose flotation instead of sedimentation?
14. A tannery treats 500 m³/d with influent TSS of 2,800 mg/L and 75% removal. Calculate the daily dry sludge production, and the wet volume at 4% solids.

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LOOKING AHEAD

Chapter 2 (Secondary Treatment) takes up the dissolved and colloidal pollutants that primary treatment cannot remove — the physicochemical and advanced processes: adsorption, chemical oxidation, ion exchange, filtration and membrane separation, including the removal of the dissolved salts that no unit in this chapter has touched.

Chapter 3 (Biological Treatment) covers the living processes that remove the dissolved biodegradable organic matter — the activated sludge process, aeration systems, trickling filters, rotating biological contactors, anaerobic reactors and pond systems.